Clinical Term Mapping

Entity + note context → ICD-10-CM code & SNOMED CT concept

wvyern | Final approach and demo

Dev (780 questions)	Test (1,025 questions)
ICD **0.573** SNOMED **0.667**	ICD **0.564** SNOMED **0.612**
shortlist recall 0.82 / 0.89	overall 0.590; headline = lower system

Two stages: judge-ranked shortlist → calibrated decision model, ensembled over option orders.

Problem, data, and what makes it hard

Task

One entity + its note context → exactly one ICD-10-CM and one SNOMED CT id.

Data

2,882 entity/chunk pairs; leak-free splits

train 1,873 | dev 432 | test 577

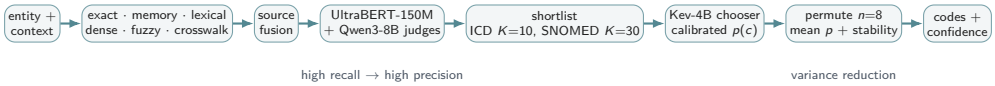
Metric

Exact match per system; **headline = lower of the two**.

Why it is hard

- **Long tail:** 32% of dev ICD and 64% of dev SNOMED golds never appear in train.
- **Conventions:** gold `E78.0` vs. billable child `E78.00`.
- **Noise:** abbreviations (*lee*), OCR artifacts (*ø tremor*).
- **Near-duplicate codes:** right concept, wrong specificity.

Architecture: retrieval, then a calibrated decision



Stage 1 — shortlist

7 candidate sources + 3 dense encoders; two judges rank candidates; keep K small to avoid near-gold decoys.

Stage 2 — chooser

Small decision model returns probabilities, not text; permutation ensemble removes order sensitivity.

Stage 1: candidate pool → shortlist

Candidates
exact · memory · lexical · dense · fuzzy · crosswalk · char-ngram
dense pool: gte-base + BioLORD + UltraBERT-Embed (weight 0.84)

Judges
UltraBERT-150M pairwise, fine-tuned on 60.6k pairs vs. 20 hard negatives
Qwen3-Reranker-8B (zero-shot)
blend: $j = 0.5\, z_u + 0.5\, z_q$

Shortlist
rank by $w\,\hat{j} + (1 - w)\,\hat{f}$, top-$K$

Configuration & recall

System	Judge	w	K
ICD	UltraBERT	0.2	10
SNOMED	combined	0.6	30

dev recall **0.818** / **0.887**
test recall (config B) 0.838 / 0.874

Larger K raises recall but seeds decoys: measured on dev, it *lowers* accuracy.

Stage 2: calibrated chooser + permutation ensemble

Kev-4B (Qwen3.5-4B-Base + LoRA + pointer head)

- Prefill-only: returns a probability for every candidate. No generation.
- Questions isolated; checkpoint ships a fitted temperature.
- Trained on 1,776 examples built with the *inference* recipe; 2 epochs, 44 min.

Why not a generative LLM?
Qwen3.5-4B instruct replacement: **0.402 / 0.408**; it re-codes clinically and ignores dataset conventions.

Permutation ensemble

$$\bar{p}(c) = \frac{1}{n} \sum_{t=1}^{n} p_t(c \mid \pi_t)$$

predict $\arg\max_c \bar{p}(c)$; record margin $\bar{p}_{(1)} - \bar{p}_{(2)}$ and stability.

n=8 option orders (seeded).

dev ICD **0.548** $\rightarrow$ **0.573**
dev SNOMED **0.647** $\rightarrow$ **0.667**

Order alone can swing accuracy $\sim$1.4 pts.

Results

Split	Config	ICD	SNOMED	All
dev	A, single	0.548	0.647	0.598
dev	A, perm. n=8	**0.573**	**0.667**	**0.620**
test	A, single	0.532	0.603	0.571
test	B, perm. n=8	**0.564**	**0.612**	**0.590**

B = ICD shortlist widened to K=20 with the combined judge (test recall 71%→84%).

Diagnostics (test)

shortlist recall 0.838 / 0.874
acc|present 0.673 / 0.700
ICD 3-char partial credit **0.748**
unseen-gold accuracy 0.507 / 0.564

unseen golds: 146 ICD / 319 SNOMED

Where it fails — and the demo cases

Entity	Failure mode	Pick / gold
ich	abbreviation	I61.9 / I61.9
lee	abbrev. collision	**A57** / I89.0
ø tremor	OCR artifact	R25.1 / R25.1
t2dm poorly controlled	specificity	E11.65 / E11.65
hypercholesteremia	convention	**E78.0** / E78.00
pulmonary nodule	sibling near-miss	**R91.8** / R91.1

demo: `python demo/inspect.py` shows top-3 probabilities, margin, stability.

Coverage / abstention (test)

coverage	ICD	SNOMED
100%	0.564	0.612
70%	0.679	0.740
60%	**0.721**	**0.770**
50%	0.761	0.809

rank by ensemble margin

Error split: ICD 66 retrieval / 89 chooser; SNOMED 47 / 92.

61 of 155 ICD errors are same-block siblings.

What did not work, and what is next

Measured and rejected

- × Larger K on dev (0.521 / 0.548)
- × Alignment retrain on served shortlists (0.532 / 0.609)
- × Generative chooser (0.402 / 0.408)
- × Gemma-4 arbitration (harmful / neutral)
- × Imported fine-tuned SapBERT+CE ICD stack (0.556–0.588)
- × Longer criteria / full context / hierarchy-first (no change)

Next

- → Parent/child *convention* tie-break learned from train labels
- → Evidence-aware arbitration only on low-margin rows
- → Kev-9B/27B (drop-in; blocked by shared GPUs)
- → ICD recall: sibling expansion + recall-oriented judge
- → Ship calibrated abstention (60% at ∼0.72–0.77)

Artifacts

Repo `hilabs-submission` | chooser https://huggingface.co/d3let3/kev-4b-clinical-mapping | judge https://huggingface.co/d3let3/ultrabert-clinical-judge
Demo video: TBD